Aditya Dendukuri

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Research Interests

Numerical analysis and scientific computing, with an emphasis on multigrid, Krylov methods, and randomized numerical linear algebra. Applications to stochastic dynamical systems, inverse problems, and scientific machine learning.

Education

University of California, Santa Barbara Sept. 2021 – Present
PhD in Computer Science, expected spring 2027. Advisor: Linda R. Petzold.

University of Arkansas, Fayetteville Aug. 2018 – July 2020
MS in Computer Science. Advisors: David M. Ford and Matthew J. Patitz.
Thesis: *Nonlinear Dimensionality Reduction for the Thermodynamics of Small Clusters of Particles*.

University of Central Arkansas Aug. 2014 – May 2018
BS in Computer Science, minor in Physics. Advisor: Tansel Halic.

Research Experience

University of California, Santa Barbara Santa Barbara, CA
Graduate Research Assistant; advisor: Linda R. Petzold Sept. 2021 – Present

- Developed an adaptive method for stiff chemical master equations that uses probability flux to select molecular states and time steps. Derived bounds for the errors introduced by truncation and numerical propagation. Published in *Multiscale Modeling & Simulation* (2026).
- Constructed multigrid transfer operators and coarse equations for the reaction–diffusion master equation. Analyzed the error introduced by coarsening and computed transient distributions with up to 7.16 million states.
- Developed adjoint Krylov methods that reuse forward and adjoint approximations to compute derivatives with respect to many unknown coefficients. Applied the methods to chemical master equations, Kuramoto–Sivashinsky dynamics, and Lindblad equations.
- Analyzed the bias introduced by projection in Koopman operator inference. Recovered ten reaction rates from a 15×15 operator for a model with more than 146 million molecular states.
- Contributed analyses of time series and reduced models for neuronal firing sequences in human brain organoids (*Nature Neuroscience*, 2026).

Lawrence Berkeley National Laboratory Berkeley, CA
UC Graduate Student Researcher Intern; mentor: Mark Fornace July – September 2026

- Developed methods that accelerate the simulation of continuous-time Markov chains by replacing repeated transitions within a subnetwork with a calculation of where the process exits.
- Implemented subnetwork sweeping in C++, using sparse factorizations and estimates of occupation times to guide the calculation. A public release is planned.

NSF National Center for Atmospheric Research Boulder, CO
SIParCS Graduate Student Intern May – July 2024

- Developed C++ components of the TUV-x radiative-transfer code, including a tridiagonal solver. Compared the solver with LAPACKE and added numerical tests and performance benchmarks.

University of Arkansas Fayetteville, AR
Research Associate Sept. 2020 – June 2021
Graduate Research Assistant Aug. 2018 – July 2020

Developed Monte Carlo software in C for nucleation studies. Used diffusion maps and neural networks to identify reduced variables for the thermodynamics of small particle systems. Also studied quantum learning models.

University of Central Arkansas Conway, AR
Undergraduate Research Assistant Aug. 2016 – May 2018

Developed fluid-flow and heat-transfer models for real-time surgical training simulation.

Publications and Preprints

Published

1. **A. Dendukuri**, S. Chandrasekaran, and L. R. Petzold. **Flux-Preserving Adaptive Finite State Projection for Multiscale Stochastic Reaction Networks.** *Multiscale Modeling & Simulation* 24(4), 1664–1696 (2026). doi:10.1137/25M1823623.
2. T. van der Molen et al. **Preconfigured neuronal firing sequences in human brain organoids.** *Nature Neuroscience* 29, 123–135 (2026). doi:10.1038/s41593-025-02111-0.
3. **A. Dendukuri**, M. Tunc, D. Demirel, S. Kockara, and T. Halic. **Unified Framework for Real-Time Fluid Simulation in Virtual Rotator Cuff Arthroscopic Skill Trainer (ViRCAST).** *IEEE BIBE*, 346–353 (2023). doi:10.1109/BIBE60311.2023.00063. **Best Paper Award.**
4. D. M. Ford, **A. Dendukuri**, G. Kalyoncu, K. Luu, and M. J. Patitz. **Machine learning to identify variables in thermodynamically small systems.** *Computers & Chemical Engineering* 141, 106989 (2020). doi:10.1016/j.compchemeng.2020.106989.
5. D. Demirel, A. Yu, S. Cooper-Baer, **A. Dendukuri**, T. Halic, S. Kockara, N. Kockara, and S. Ahmadi. **A hierarchical task analysis of shoulder arthroscopy for a virtual arthroscopic tear diagnosis and evaluation platform (VATDEP).** *International Journal of Medical Robotics and Computer Assisted Surgery* 13(3), e1799 (2017). doi:10.1002/rcs.1799.

Under review

6. **A. Dendukuri**, S. Chandrasekaran, and L. R. Petzold. **An Adjoint Krylov Approach to Parameter Inference for Affine Families of Linear Evolution Equations.** Under review, *SIAM Journal on Scientific Computing* (2026). doi:10.48550/arXiv.2610.15940.
7. **A. Dendukuri** and L. R. Petzold. **Koopman Operator Inference for the Chemical Master Equation.** Under review, *SIAM Journal on Applied Dynamical Systems* (2026). doi:10.48550/arXiv.2610.15937.
8. **A. Dendukuri** and L. Petzold. **A Multigrid Finite State Projection Algorithm for the Solution of the Reaction-Diffusion Master Equation.** Under review, *SIAM Journal on Scientific Computing*, Iterative and Multigrid Special Section (2026).

Manuscripts in preparation

- **A. Dendukuri** and M. Fornace. **Subnetwork sweeping for continuous-time Markov chains on discrete state spaces.**

Teaching Experience

University of California, Santa Barbara Sept. 2021 – Present
Graduate Teaching Assistant

- Extensive teaching assistant experience in CS111, *Introduction to Computational Science*. The course covers numerical algorithms and the use of mathematical software in scientific computation.
- Additional teaching assistant experience in discrete mathematics. Teaching experience also includes numerical methods and data structures, using C++, Python, and MATLAB.

Teaching Materials

Scientific Computing notes. Public notes that began as a study guide for CS111 and developed into a working reference. The notes connect definitions and proofs to numerical examples, algorithms, and exercises. Topics include numerical linear algebra, interpolation and quadrature, iterative solvers, ordinary differential equations, and applications to stochastic systems and inference.

Software

- **numerics**. C++20 library for scientific computing, including sparse and structured factorizations, Krylov methods, and matrix functions. Supports reuse of factorizations and LU factorization in mixed precision, with numerical tests and performance benchmarks.
- **MarkovKit**. C++20 library implementing adaptive finite state projection, stochastic simulation, and multigrid solvers for spatial reaction systems.
- **AdjointKrylovGradients**. Julia package for computing gradients through adjoint Krylov approximations. An `rrule` interface allows the method to be used in automatic differentiation. Includes code for reproducing the numerical experiments.
- **approxchol**. Header-only C++ implementation of randomized approximate Cholesky factorization for graph Laplacians.

Selected Conference Presentations

- **Data-Driven Rate Recovery and Sensitivity Analysis for Discrete Stochastic Chemical Reaction Systems.** SIAM Annual Meeting, Cleveland, OH (2026).
- **Adaptive Multigrid Finite State Projection for the Reaction-Diffusion Master Equation.** Colorado Conference for Iterative and Multigrid Methods, Boulder, CO (2026).
- **Detecting Subspaces for Small Lennard-Jones Clusters Using Spectral Graph Theory and Machine Learning.** SIAM Conference on Computational Science and Engineering (2021).

Professional Service

Session chair, CP17: Stochastic Systems, SIAM Annual Meeting, Cleveland, OH (2026).

Honors

- Best Paper Award, IEEE Bioinformatics and Bioengineering Conference (2023).
- Master's thesis nominated for the Conference of Southern Graduate Schools Outstanding Master's Thesis Award.
- Graduate Student Council Travel Grant, FOPAM (2019); Advancement of Undergraduate Research in the Sciences research funding (2016).